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ORAL RECEPTACLE

Abstract

An oral receptacle (10) having a receptacle body (12). The receptacle body (12) contains an effervescent components (14) comprising an acid (16) and a carbonate (18) and a medicinal and/or psychoactive component (20) for producing a medicinal and/or psychoactive effect on a user, wherein the receptacle body (12) is configured so that when the receptacle body (12) is received in the user's mouth, the effervescent components (14) forms a foam which distributes the medicinal and/or psychoactive component (20).

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a Bypass Continuation in Part of PCT International Application No. PCT/EP2025/061763 filed on Apr. 29, 2025, which claims priority to PCT International Application No. PCT/EP2024/051438 filed on Jan. 22, 2024, both of which are hereby expressly incorporated by reference, in their entirety, into the present application.

BACKGROUND

1. Field of the Disclosure

[0002] The present invention relates to an oral receptacle.

2. Description of Background

[0003] Consuming or inhaling some medicinal or psychoactive compounds such as nicotine, cannabidiol (CBD), or tetrahydrocannabinol (THC) may conventionally rely on burning substances, for example burning tobacco in cigarettes, and inhaling the smoke produced.

[0004] Electronic cigarettes (e-cigarettes) are an alternative to this, in which electrical energy is used to produce a vapour to be inhaled, instead of smoke.

[0005] The inhalation of the smoke or vapour may have medicinal effects on the user, or psychoactive effects such as stimulatory or relaxing effects. The effects produced are due to the chemicals contained within the cigarette or e-cigarette, for example nicotine, CBD or THC.

[0006] As well as the stimulating effects from the chemicals, the smoke or vapour can provide a physical sensation in the user's mouth.

[0007] There are well known health concerns regarding smoke inhalation from cigarettes, including lung and throat cancer. People often seek alternative ways to consume the chemicals which are less hazardous than inhaling smoke.

[0008] Oral nicotine pouches and tobacco pouches, such as snus, are a common alternative, the contents of which are absorbed through the oral mucosa. However, prolonged use of these pouches can result in the risk of oral injury from repeated placement of the said pouches in the same place.

[0009] Whilst e-cigarettes are considered to pose fewer health concerns, the requirement to provide an electrical energy supply can be inconvenient, requiring frequent recharging or replacement of batteries or devices, along with associated environmental concerns. The user is also often not able to accurately gauge how much nicotine they have consumed per day due to a given quantity of liquid being used to produce vapour over a long duration.

[0010] It would therefore be desirable to provide a way of delivering medicinal and/or psychoactive components, accompanied by a mouth sensation, without conventional associated production of smoke or vapour.

SUMMARY OF THE DISCLOSURE

[0011] It is an object of the present invention to reduce or substantially obviate the above problems.

[0012] According to a first aspect of the invention, there is provided an oral receptacle comprising: a receptacle body for being received in a user's mouth; an effervescent means comprising an acid and a carbonate located within the receptacle body; and a medicinal and/or psychoactive component for producing a medicinal and/or psychoactive effect on a user, the medicinal and/or psychoactive component being located within the receptacle body, the effervescent means for forming a foam in the user's mouth which distributes the medicinal and/or psychoactive component.

[0013] Harmful inhalation of smoke is avoided, and the user is still able to experience a medicinal

and/or psychoactive effect. The receptacle body can be placed in the user's mouth to release the carbonate and acid within, which can then mix with and/or dissolve in the user's saliva to react to produce a foam via generation of carbon dioxide. The foam covers a large surface area in the user's mouth and thus increases the rate of absorption of the medicinal and/or psychoactive component by the oral mucosa of the user's mouth. The receptacle is therefore configured to provide buccal or sublingual absorption of an active ingredient. This is known to be the fastest route to the brain of the active ingredient and thus the medicinal and/or psychoactive effect is achieved quickly after the foam is absorbed. The foam also allows for an enhanced physical mouth experience.

[0014] Preferably, the receptacle body or at least part of the effervescent means is provided by a liquid receptacle containing liquid. This provides additional liquid, in case the user's saliva is insufficient to fully mix and/or dissolve the acid and carbonate and/or the medicinal and/or psychoactive components. The liquid receptacle may be provided in an outer receptacle body, or there may be no separate outer receptacle body, with the liquid receptacle simply defining the receptacle body.

[0015] Optionally, the liquid receptacle is flexible. The user can apply pressure to the liquid receptacle to cause it to burst and release the liquid within.

[0016] Preferably, the liquid receptacle is frangible. Following breakage of the liquid receptacle, the liquid can be released into its surroundings to dissolve the acid and carbonate and/or medicinal and/or psychoactive component to form the foam.

[0017] Beneficially, the liquid receptacle may comprise at least one liquid-transfer conduit between the liquid receptacle and the acid and the carbonate and/or the medicinal and/or psychoactive component. The release of liquid, upon perforation of the liquid receptacle, is directed into the acid and the carbonate to more effectively produce the foam.

[0018] Optionally, the liquid-transfer conduit has a seal which is frangible when pressure is applied to the liquid receptacle. The frangible seal is an area of weakness in the liquid receptacle and promotes perforation at the seal when pressure is applied. This directs the liquid to the acid and the carbonate to more efficiently produce the foam.

[0019] Preferably, the receptacle body or at least part of the effervescent means may be provided by a semi-solid substance. A semi-solid substance, such as a gel, paste or cream, may be simpler to manufacture than a liquid receptacle.

[0020] Advantageously, the gel or gelatinous substance may be devoid or substantially devoid of water. Since the carbonate and acid may be soluble in water, water should be avoided in the semi-solid substance.

[0021] In a preferable embodiment, the semi-solid substance may be a gel or gelatinous substance. Whilst a gel is preferred, other gel-like substances may also be considered with a similar viscosity or texture. The gel may be provided in an outer receptacle body, or there may be no separate outer receptacle body, with the gel simply defining the receptacle body. The effervescent means may be mixed or dissolved into the gel.

[0022] Optionally, the gel or gelatinous substance may be an oleogel.

[0023] Preferably, the semi-solid substance may be a paste.

[0024] Beneficially, at least part of the effervescent means is provided as a powder. This provides a higher surface area for the liquid and/or saliva to efficiently produce the foam. The powders may be in compressed form, such as a tablet. Alternatively, the powder may be mixed into the semi-solid substance. Whilst a powder is preferred, it will be appreciated that other solid forms may be considered. Alternatively, one or both of the acid and carbonate may be provided in liquid form.

[0025] Preferably, at least part of said powder coats the semi-solid substance or liquid receptacle. By coating the semi-solid substance or liquid receptacle, effervescence may commence as soon as the semi-solid substance or liquid receptacle comes into contact with the user's mouth. As such, this effervescence may assist with breaking down the semi-solid substance or liquid receptacle.

[0026] Advantageously, further comprising a stabiliser or surfactant for stabilising the foam, the

stabiliser or surfactant located within the receptacle body. The stabiliser or surfactant may comprise polysorbate **20**, hydroxypropyl methylcellulose and/or lecithin. The stabiliser or surfactant maintains the structure of the foam in the user's mouth and so may provide a longer lasting mouth sensation.

[0027] Preferably, the medicinal and/or psychoactive component is one or more of nicotine, caffeine, cannabidiol, tetrahydrocannabinol, a vitamin, a pharmaceutical, and/or alcohol.

[0028] Preferably, the receptacle body comprises at least any one of plant fibres, hydrophilic polymers, and gelatine. The plant fibres may be bound by a resin. The plant fibres, and/or resin where present, may be food safe and/or digestible, so that the entire receptacle may be consumed leaving no waste product to discard. The hydrophilic polymers may allow the receptacle body to be instantly dissolvable.

[0029] Advantageously, the receptacle body may be flexible for ease of manipulation in a mouth of the user. The flexibility of the receptacle body may mean that pressure can be applied through the receptacle body to the liquid receptacle and therefore allows for the receptacle body and liquid receptacle to be burst.

[0030] Preferably, the receptacle body is frangible. The user can perforate the receptacle body with their teeth or gums to access the components within.

[0031] Beneficially, the acid may comprise citric acid. Preferably, the carbonate comprises sodium carbonate or sodium bicarbonate. The citric acid and sodium carbonate or bicarbonate are food-safe and easily accessible compounds. The reaction of the acid and the carbonate, preferably via liquid, causes the production of carbon dioxide which generates the foam.

[0032] Preferably, the receptacle body is soluble in water. This allows for the receptacle body to dissolve when placed in the mouth due to the presence of saliva. This causes the release of the substances within the receptacle body without necessarily requiring pressure to be applied to the oral receptacle.

[0033] Optionally, the receptacle body is insoluble in water. This gives the user more control on when to activate the receptacle in the mouth. The user can apply pressure to perforate the receptacle to release the substances within the receptacle body.

[0034] Preferably, the receptacle body comprises at least one fin. The fins allow the user to easily move and manipulate the receptacle within their mouth to position the receptacle correctly.

[0035] Advantageously, the effervescent means and/or the medicinal and/or psychoactive component may be defined by a plurality of spaced apart components which are distributed within the receptacle body. Optionally, there are at least ten said components. Having a larger number of components provides a higher surface area for the effervescent means and the medicinal and/or psychoactive components to mix such that the medicinal and/or psychoactive components are distributed evenly throughout the foam produced. The components may be capsules, balls or beads of the effervescent means and/or the medicinal and/or psychoactive component.

[0036] Preferably, the oral receptacle further comprises a flavouring for causing increased saliva production, such as menthol. Additional saliva may assist with breaking down the receptacle body, semi-solid substance and/or liquid receptacle, and/or may assist with foam production.

[0037] Preferably, the oral receptacle is a tobacco industry product. The oral receptacle is therefore preferably, but not exclusively, a non-smoke substitute for conventional tobacco products such as cigarettes, and e-cigarettes. The oral receptacle is also a substitute for nicotine pouches and tobacco pouches; removing the risk of the user having oral injuries.

[0038] According to a second aspect of the invention, there is provided an oral receptacle comprising: a receptacle body for being received in a user's mouth; an effervescent means located within the receptacle body; and a medicinal and/or psychoactive component for producing a medicinal and/or psychoactive effect on a user, the medicinal and/or psychoactive component being located within the receptacle body, the effervescent means for forming a foam in the user's mouth which distributes the medicinal and/or psychoactive component.

[0039] According to a third aspect of the invention there is provided an oral receptacle, according to the first or second aspects, for use in buccally and/or sublingually delivering a medicinal and/or psychoactive component.

[0040] According to a fourth aspect of the invention, there is provided a method of buccal and/or sublingual delivery of a medicinal and/or psychoactive component using the oral receptacle according to the first, second or third aspects.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0041] Various aspects of at least one embodiment are discussed below with reference to the accompanying figures, which are not intended to be drawn to scale. The figures are included to provide illustration and a further understanding of the various aspects and embodiments, and are incorporated in and constitute a part of this specification, but are not intended as a definition of the limits of the disclosure. In the figures, each identical or nearly identical component that is illustrated in various figures is represented by a like numeral. For purposes of clarity, not every component may be labelled in every figure. In the figures:

[0042] FIG. 1 is a perspective view of a first embodiment of an oral receptacle in accordance with first, second, third and fourth aspects of the invention containing a liquid receptacle;

[0043] FIG. 2 is a front view of a second embodiment of the oral receptacle in accordance with the first, second, third and fourth aspects of the invention containing the liquid receptacle;

[0044] FIG. 3 is a perspective view of the second embodiment of the oral receptacle of FIG. 2;

[0045] FIG. 4 is a side view of the second embodiment of the oral receptacle of FIG. 2;

[0046] FIG. 5 is a perspective view of a third embodiment of the oral receptacle in accordance with the first, second, third and fourth aspects of the invention without a liquid receptacle;

[0047] FIG. 6 is a perspective view of a fourth embodiment of the oral receptacle in accordance with the first, second, third and fourth aspects of the invention showing a plurality of spaced apart components;

[0048] FIG. 7 is a front view of a fifth embodiment of the oral receptacle in accordance with the first, second, third and fourth aspects of the invention; and

[0049] FIG. 8 shows a diagrammatic representation of a front view of a sixth embodiment of the oral receptacle in accordance with the first, second, third and fourth aspects of the invention.

DETAILED DESCRIPTION

[0050] Referring to FIG. 1, there is shown a first embodiment of an oral receptacle **10** having a receptacle body **12**, an effervescent means **14** comprising an acid **16** and a carbonate **18**, and a medicinal and/or psychoactive component **20**. The oral receptacle **10** is a tobacco industry product.

[0051] The receptacle body **12** is a case or pouch. In FIG. 1, the case is illustrated as being transparent to allow the effervescent means **14** and medicinal and/or psychoactive component **20** to be seen within. However, the case may not necessarily be transparent, and may be opaque or translucent.

[0052] The receptacle body **12** is also insoluble in water and perforable so that when the user applies pressure to the receptacle body **12**, the components within the receptacle body **12** are released. The receptacle body **12** also comprises plant fibres, for example hemp or cellulose. The plant fibres are preferably edible and/or digestible. It can be appreciated that the receptacle body **12** may be soluble in water and perforable.

[0053] The receptacle body **12** is flexible for ease of manipulation in the user's mouth so that the user can fit the receptacle body **12** between their gums or teeth. Additionally, the flexibility allows pressure to be applied by the user through the receptacle body **12** to a liquid receptacle **22** within to burst both the receptacle body **12** and the liquid receptacle **22**. The liquid receptacle **22** containing

liquid may also form part of the effervescent means.

[0054] The receptacle body **12** comprises at least one fin **24** as shown more clearly in FIG. **3**. The fin **24** is located substantially around a perimeter of the receptacle body **12** and extends outwards from the receptacle body **12**. The fin **24** provides the user with a surface to more easily move the oral receptacle **10** around their mouth.

[0055] The medicinal and/or psychoactive component **20** is one or more of nicotine, caffeine, cannabidiol, tetrahydrocannabinol, a vitamin, a pharmaceutical and/or alcohol. The medicinal and/or psychoactive component **20** is in the form of a powder, although in an alternative embodiment the medicinal and/or psychoactive component **20** may be in liquid form. The medicinal and/or psychoactive component **20** provides relief, stimulation or other medicinal and/or psychoactive effect on the user upon absorption. The medicinal and/or psychoactive component **20** is located within the receptacle body **12**.

[0056] The effervescent means **14** is located within the receptacle body **12**. The acid **16** and the carbonate **18** are powders and in this embodiment are citric acid and sodium carbonate or sodium bicarbonate. The acid **16** and the carbonate **18** and the medicinal and/or psychoactive component **20** may be in the form of a tablet **26** or pill. The carbonate **18**, acid **16**, medicinal and/or psychoactive component **20** may be intermixed in the or each tablet **26** or pill. There are shown two such tablets **26** in FIG. **1** for provision and distribution of a larger amount of the medicinal and/or psychoactive component **20**.

[0057] The liquid receptacle **22** is located within the receptacle body **12** and disposed between the tablets **26**. The liquid receptacle **22** is a further component of the effervescent means **14**, and contains liquid. The liquid is preferably water or any other edible liquid. The liquid receptacle **22** is flexible and also perforable such that the liquid can be released to mix with the acid **16** and carbonate **18** to form a foam. The liquid receptacle **22** is a pill case containing liquid and may be ovoid in shape.

[0058] The liquid comprises a stabiliser or surfactant, for example polysorbate **20**—also known as Tween (RTM) **20**—or lecithin. The stabiliser stabilises the foam produced from the mixing of the liquid and the acid **16** and carbonate **18**, and maintains the structure of the foam until the user ingests it. However, it will be appreciated that the stabiliser or surfactant may be provided in a solid form and incorporated into the tablet.

[0059] In use, the user inserts the oral receptacle **10** into their mouth and bites down on the receptacle body **12**. The receptacle body **12** is configured so that when pressure is applied by the user's teeth or gums, the pressure is also applied to the liquid receptacle **22** which causes the receptacle body **12** and the liquid receptacle **22** to burst, releasing the liquid.

[0060] The released liquid mixes with and dissolves the acid **16** and carbonate **18** to form the foam. The foam combines with the medicinal and/or psychoactive components **20** to form a medicinal and/or psychoactive foam which is distributed throughout the user's mouth. The foam has a large surface area in the user's mouth, and therefore the active ingredient of the medicinal and/or psychoactive component **20** may be easily absorbed by the user through the oral mucosa.

[0061] Referring to FIGS. **2** to **4**, there is shown a second embodiment of the oral receptacle **110**. Similar features to the first embodiment have similar reference numerals with **100** added. Further detailed description has been omitted for brevity.

[0062] The oral receptacle **110** comprises the receptacle body **112**, the effervescent means **114** comprising the acid **116**, the carbonate **118**, and the liquid receptacle **122**, and the medicinal and/or psychoactive component **120**.

[0063] The fins **124** are shown more clearly in FIG. **3** outlines by the solid black lines.

[0064] Two liquid-transfer conduits **128** extend from the liquid receptacle **122** to the tablets **126** comprising the acid **116** and the carbonate **118** and/or the medicinal and/or psychoactive component **120**. The liquid-transfer conduits **128** are best seen in FIG. **2**, and may be termed tubes.

[0065] The liquid-transfer conduits **128** may each have a seal **130** which is frangible when pressure

is applied to the liquid receptacle **122**. The seal **130** may be located at the interface between the liquid-transfer conduit **128** and the tablet **126**. The seal **130** provides a weakness, or predetermined breaking point, at which the liquid receptacle **122** breaks or perforates and allows for the liquid to be directed into the tablets **126**. This arrangement efficiently produces the medicinal and/or psychoactive foam as the liquid is focussed directly onto the tablets **126** and thus all of the liquid dissolves the acid **116** and carbonate **118** and medicinal and/or psychoactive components **120**. Although the term seal is used, it will be appreciated that this may in fact only be a portion of thinner wall of the liquid receptacle **122**, or an otherwise weakened portion, which therefore allows this area to break in preference of the remainder of the liquid receptacle.

[0066] FIG. **4** shows that a thickness of the liquid receptacle **122** is thicker than a thickness of the tablet **126**. The location of the liquid-transfer conduit **128** is located in a central portion of the liquid receptacle **122**, the position of the liquid-transfer conduit **128** shown by a dashed line. It is also shown that a cross section of the receptacle body **112** and a cross section of the liquid receptacle **122** is substantially elliptical.

[0067] The mechanism of use is the same as the first embodiment, except that the liquid is directly delivered to the tablets **126** via the liquid-transfer conduits **128** upon perforation of the receptacle body **112** and liquid receptacle **122**.

[0068] Referring to FIG. **5**, there is shown a third embodiment of the oral receptacle **210**. Similar features to the first embodiment have similar reference numerals with **200** added. Further detailed description has been omitted for brevity.

[0069] The liquid receptacle is omitted in the third embodiment. The oral receptacle **210** comprises the receptacle body **212**, the effervescent means **214** comprising the acid **216** and the carbonate **218**, and the medicinal and/or psychoactive component **220**.

[0070] The effervescent means **214** in this embodiment only comprises the acid **216** and the carbonate **218**.

[0071] In use, the user inserts the oral receptacle **210** into their mouth and bites down on the receptacle body **212**. The receptacle body **212** is configured so that when pressure is applied by the user's teeth or gums, the receptacle body **212** bursts.

[0072] The exposed acid **216** and carbonate **218** dissolve and mix with the user's saliva to form the foam. The foam combines with the medicinal and/or psychoactive components **220** to form a medicinal and/or psychoactive foam which is distributed throughout the user's mouth. The medicinal and/or psychoactive component **220** is easily absorbed by the user.

[0073] Referring to FIG. **6**, there is shown a fourth embodiment of the oral receptacle **310**. Similar features to the first embodiment have similar reference numerals with **300** added. Further detailed description has been omitted for brevity.

[0074] The liquid receptacle is omitted in the fourth embodiment. The oral receptacle **310** comprises the receptacle body **312**, the effervescent means **314** comprising the acid **316** and the carbonate **318** and the medicinal and/or psychoactive component **320**.

[0075] The receptacle body **312** comprises a matrix **332** in which the effervescent means **314** and medicinal and/or psychoactive component **320** are distributed or suspended. The matrix **332** comprises the same material as the receptacle body **312**.

[0076] The effervescent means **314** and the medicinal and/or psychoactive component **320** are defined by a plurality of spaced apart components **334** which are distributed within the matrix **332** of the receptacle body **312**. There are shown twenty-two components **334** in FIG. **6**, and thus there are more than ten components **334** within the receptacle body **12**. Each of the plurality of components **334** has a smaller surface area compared to the surface area of the tablets **26** of the previous embodiments.

[0077] In use, the user inserts the oral receptacle **310** into their mouth. The receptacle body **312** is soluble in water in this embodiment and thus dissolves in the user's mouth when in contact with saliva. It can be appreciated that the receptacle body **312** can be insoluble in water. Pressure may be

applied by the user to speed up the process.

[0078] Once the receptacle body **312** is dissolved, the exposed acid **316** and carbonate **318** dissolve and mix with the user's saliva to form the foam. The foam combines with the medicinal and/or psychoactive components **320** to form a medicinal and/or psychoactive foam which is distributed throughout the user's mouth. The foam, and therefore the medicinal and/or psychoactive component **320** is easily absorbed by the user.

[0079] Referring to FIG. 7, there is shown a fifth embodiment of the oral receptacle **410**. Similar features to the first embodiment have similar reference numerals with **400** added. Further detailed description has been omitted for brevity.

[0080] The fifth embodiment is similar to the fourth embodiment with the addition of a liquid receptacle **422**. The effervescent means **414** comprises the liquid receptacle **422** together with a plurality of components **434** containing acid **416** and carbonate **418**. The plurality of components **434** may also contain the medicinal and/or psychoactive components **420**.

[0081] The liquid receptacle **422** may include a liquid-transfer conduit **428** or protrusion, which is preferably tubular. The liquid-transfer conduit **428** or protrusion may extend from the liquid receptacle **422** into the matrix **432** and may direct liquid into the matrix **432**.

[0082] The liquid receptacle **422** may include at least one seal **430**, or predetermined breaking point, which is frangible to allow emission of liquid when the liquid inside the liquid receptacle is pressurised. The liquid may be pressurised via compression or deformation of the liquid receptacle **422**.

[0083] The receptacle body **412** is insoluble in water in this embodiment, and the liquid may be dispensed therefrom only once the receptacle body **412** is bitten through. However, it is envisaged that the receptacle body **412** could be soluble.

[0084] In use, the user inserts the oral receptacle **410** into their mouth and bites down on the receptacle body **412**. The oral receptacle **410** may be punctured or broken by this biting.

[0085] When the receptacle body **412** is bitten through and/or deformed by the user's teeth or gums, pressure is applied to the liquid receptacle **422** which causes the receptacle body **412** and the liquid receptacle **422** to burst, preferably at the pre-determined breaking points, releasing the liquid.

[0086] The liquid is dispersed from the burst liquid receptacle **422** to the plurality of components **434** via the liquid-transfer conduits **428**.

[0087] The released liquid mixes with and dissolves the acid **416** and carbonate **418** to react to form carbon dioxide which forms the foam. The foam includes the medicinal and/or psychoactive components **420** forming a medicinal and/or psychoactive foam which is distributed throughout the user's mouth. The foam, and therefore the medicinal and/or psychoactive component **420** is easily absorbed by the user.

[0088] Referring now to FIG. 8, there is shown a sixth embodiment of an oral receptacle **510**. Similar or identical reference numerals as per the first embodiment are used with the sixth embodiment, with **500** added.

[0089] Instead of a liquid receptacle, as per some of the preceding embodiments, a gel **522** is used. The gel **522** may be a firm gel body, or may be a softer gel which has a skin or casing.

[0090] Preferably the gel **522** is devoid of water, or substantially devoid of water. In other words, the gel **522** is an anhydrous gel. More preferably, the gel **522** is an oil-based gel, such as an oleogel and/or a lipophilic matrix. Any suitable food-safe oil may be used for preparation of the oil-based gel, but medium-chain triglycerides are preferred.

[0091] Whilst an oil is preferred, other bases may be used for the gel **522**, provided it is not a solvent for, nor reactionary with, the effervescent means **514**.

[0092] The gel **522** may be formed by a gel former such as fumed silica, for example Aerosil (RTM).

[0093] At least part of the effervescent means **514** is provided by the gel **522**. In this case, at least part of the effervescent means **514** may be mixed into the gel **522**. Here the carbonate and the acid

are provided in a powdered form, and the powder is mixed into the gel 522. The powder is represented in FIG. 8 as small circles within the gel 522. Since the gel 522 is anhydrous, the powder does not dissolve into the gel 522 and so the acid and carbonate do not react. The carbonates and acids may be of the type discussed for preceding embodiments.

[0094] Alternatively, it will be appreciated that one of the acid and carbonate may be dissolved into the gel 522, with the other of the acid and carbonate insoluble in the gel 522 and mixed through.

[0095] At least part of the effervescent means is provided as a coating or covering 536 of the gel 522. In this case, the carbonate and acid, provided in powdered form, are adhered to the gel 522 to provide a coating 536.

[0096] The gel 522 preferably comprises a surfactant for production of a foam when the effervescent takes place. The surfactant is preferably polysorbate 20.

[0097] Additionally, the gel 522 preferably includes a stabiliser or gel thickener for stabilising or thickening the gel 522. This may be hydroxypropyl methylcellulose.

[0098] A permeation enhancer may be included, such as sodium dodecyl sulfate.

[0099] The gel 522 also includes the active ingredient, which is the medicinal and/or psychoactive component, such as any of those described for preceding embodiments.

[0100] The receptacle body 512, or bag, is preferably dissolvable, and more preferably instantaneously dissolvable, and so may comprise or consist essentially of hydrophilic polymers. However, the receptacle body may not necessarily be dissolvable and/or may comprise other materials such as plant fibres and/or gelatine.

[0101] A flavouring, such as menthol, may also be provided to increase saliva production. The flavouring may be provided in the receptacle body 512 or in or on the gel 522.

[0102] An example composition of the oleogel is as follows:

TABLE-US-00001 Component Weight percentage Citric Acid 11 Sodium bicarbonate 16 Polysorbate 20 1 Hydroxypropyl methylcellulose 0.2 Sodium dodecyl sulfate 0.1 Fumed silica 5 Flavouring Variable Active Ingredient Variable Medium-chain triglycerides Balance (e.g. around 70)

[0103] The oleogel may be prepared by putting the fumed silica in a mortar and then adding the medium-chain triglycerides while constantly stirring the mixture using a pestle. The fumed silica should ideally be prevented from clotting and incorporation of air into the mixture should be avoided. Water should not be included.

[0104] The sixth embodiment of the oral receptacle 510 is used in a similar way as the preceding embodiments. The oral receptacle 510 is placed in the user's mouth, resulting in the receptacle body 512 dissolving in the user's saliva.

[0105] The user's saliva then contacts the effervescent means coating 536 on the gel 522, which begins to effervesce due to the reaction between the acid and carbonate and/or produces a foam. This may assist with breaking down the gel 522, producing further effervescence, generating the stable foam. The stable foam contains the active ingredient, which can then persist in the user's mouth and be absorbed buccally and/or sublingually.

[0106] In any of the aforementioned embodiments, the receptacle body may have a width of between 30 mm and 35 mm, for example 32 mm and 35 mm, a height of 15 mm and a thickness of 8 mm. However, it is appreciated that the receptacle body may have any other combination of dimensions suitable to fit within a user's mouth.

[0107] The liquid receptacle, if present, may have a diameter of 6.5 mm and a height of 12 mm. It is appreciated that the liquid receptacle may have any other combination of dimensions suitable to fit within the receptacle body.

[0108] The or each tablet, if present, may have a height of 13 mm and a width of 10 mm, although other dimensions may be considered to fit within the receptacle body.

[0109] In an alternative embodiment, the receptacle body comprises gelatine.

[0110] In an alternative embodiment, the acid comprises acetic acid, tartaric acid, malic acid, folic

acid, fumaric acid, lactic acid or other edible acid.

[0111] In an alternative embodiment, the carbonate may comprise any other edible carbonate compound, such as calcium carbonate.

[0112] In an alternative embodiment, there may be one or more than two tablets present in the receptacle body.

[0113] In an alternative embodiment, the acid and carbonate and the medicinal and/or psychoactive component may be spaced apart from one another in different tablets.

[0114] In an alternative embodiment, the tablet may be a receptacle containing the acid and carbonate and/or the medicinal and/or psychoactive component in solid form. The receptacle may be a flexible pill or receptacle similar to the receptacle body and liquid receptacle such that pressure can be applied to burst the receptacle and release the acid and carbonate and/or the medicinal and/or psychoactive component to be dissolved in liquid and/or saliva.

[0115] In an alternative embodiment, the acid and the carbonate and the medicinal and/or psychoactive component may be in solution.

[0116] In an alternative embodiment, the liquid may comprise at least one of the acid and the carbonate and the medicinal and/or psychoactive component.

[0117] In an alternative embodiment, the medicinal and/or psychoactive component may be any other medicinal and/or psychoactive substance, for example paracetamol or ibuprofen.

[0118] It is therefore possible to provide a means for consuming medicinal or psychoactive compounds without inhalation of harmful smoke, as well as distributing the compounds within the user's mouth whilst providing a pleasant mouth sensation.

[0119] The words 'comprises/comprising' and the words 'having/including' when used herein with reference to the present invention are used to specify the presence of stated features, integers, steps, or components, but do not preclude the presence or addition of one or more other features, integers, steps, components, or groups thereof.

[0120] It is appreciated that certain features of the invention, which are, for clarity, described in the context of separate embodiments, may also be provided in combination in a single embodiment. Conversely, various features of the invention which are, for brevity, described in the context of a single embodiment, may also be provided separately or in any suitable sub-combination.

[0121] The embodiments described above are provided by way of example only, and various changes and modifications will be apparent to persons skilled in the art without departing from the scope of the present invention as defined by the appended claims.

Claims

1. An oral receptacle (**10, 110, 210, 310, 410**) comprising: a receptacle body (**12, 112, 212, 312, 410**) for being received in a user's mouth; an effervescent means (**14, 114, 214, 314, 414**) comprising an acid (**16, 116, 216, 316, 416**) and a carbonate (**18, 118, 218, 318, 418**) located within the receptacle body (**12, 112, 212, 312, 412**); and a medicinal and/or psychoactive component (**20, 120, 220, 320, 420**) for producing a medicinal and/or psychoactive effect on a user, the medicinal and/or psychoactive component (**20, 120, 220, 320, 420**) being located within the receptacle body (**12, 112, 212, 312, 412**), the effervescent means (**14, 114, 214, 314, 414**) for forming a foam in the user's mouth which distributes the medicinal and/or psychoactive component (**20, 120, 220, 320, 420**).

2. An oral receptacle (**10, 110, 410**) as claimed in any one of the preceding claims, wherein the effervescent means (**14, 114, 414**) further comprises a liquid receptacle (**22, 122, 422**) containing liquid.

3. An oral receptacle (**10, 110, 410**) as claimed in claim 2, wherein the liquid contains a stabiliser for stabilising the foam.

4. An oral receptacle (**10, 110, 410**) as claimed in claim 3, wherein the stabilizer comprises

polysorbate **20**.

5. An oral receptacle (**10, 110, 410**) as claimed in claim 3 or 4, wherein the stabiliser comprises lecithin.

6. An oral receptacle (**10, 110, 410**) as claimed in any one of claims 2 to 5, wherein the liquid receptacle (**22, 122, 422**) is flexible.

7. An oral receptacle (**10, 110, 410**) as claimed in any one of claims 2 to 6, wherein the liquid receptacle (**22, 122, 422**) is frangible.

8. An oral receptacle (**110, 410**) as claimed in any one of claims 2 to 7, wherein the liquid receptacle (**122, 422**) comprises at least one liquid-transfer conduit (**128, 428**) between the liquid receptacle (**122, 422**) and the acid (**116, 416**) and the carbonate (**118, 418**) and/or the medicinal and/or psychoactive component (**120, 420**).

9. An oral receptacle (**110, 410**) as claimed in claim 8, wherein the liquid-transfer conduit (**128, 428**) has a seal (**130, 430**) which is frangible when pressure is applied to the liquid receptacle (**122, 422**).

10. An oral receptacle (**10, 110, 210, 310, 410**) as claimed in any one of the preceding claims, wherein the medicinal and/or psychoactive component (**20, 120, 220, 320, 420**) is one or more of nicotine, caffeine, cannabidiol, tetrahydrocannabinol, a vitamin, a pharmaceutical, and/or alcohol.

11. An oral receptacle (**10, 110, 210, 310, 410**) as claimed in any one of the preceding claims, wherein the receptacle body (**12, 112, 212, 312, 412**) comprises plant fibres.

12. An oral receptacle (**10, 110, 210, 310, 410**) as claimed in any one of the preceding claims, wherein the receptacle body (**12, 112, 212, 312, 412**) is flexible for ease of manipulation in a mouth of the user.

13. An oral receptacle (**10, 110, 210, 310, 410**) as claimed in any one of the preceding claims, wherein the receptacle body (**12, 112, 212, 312, 412**) is frangible.

14. An oral receptacle (**10, 110, 210, 310, 410**) as claimed in any one of the preceding claims, wherein the acid (**16, 116, 216, 316, 416**) and the carbonate (**18, 118, 218, 318, 418**) are powders.

15. An oral receptacle (**10, 110, 210, 310, 410**) as claimed in any one of the preceding claims, wherein the acid (**16, 116, 216, 316, 416**) comprises citric acid.

16. An oral receptacle (**10, 110, 210, 310, 410**) as claimed in any one of the preceding claims, wherein the carbonate (**18, 118, 218, 318, 418**) comprises sodium carbonate.

17. An oral receptacle (**10, 110, 210, 310, 410**) as claimed in any one of the preceding claims, wherein the receptacle body (**12, 112, 212, 312, 412**) is soluble in water.

18. An oral receptacle (**10, 110, 210, 410**) as claimed in any one of claims 1 to 16, wherein the receptacle body (**12, 112, 212, 412**) is insoluble in water.

19. An oral receptacle (**10, 110, 210, 310, 410**) as claimed in any one of the preceding claims, wherein the receptacle body (**12, 112, 212, 312, 412**) comprises at least one fin (**24, 124**).

20. An oral receptacle (**310, 410**) as claimed in any one of the preceding claims, wherein the effervescent means (**314, 414**) and/or the medicinal and/or psychoactive component (**320, 420**) are defined by a plurality of spaced apart components (**334, 434**) which are distributed within the receptacle body (**312, 412**).

21. An oral receptacle (**310, 410**) as claimed in claim 20, wherein there are at least ten said components (**334, 434**).

22. An oral receptacle (**10, 110, 210, 310, 410**) as claimed in any one of the preceding claims, wherein the oral receptacle (**10, 110, 210, 310, 410**) is a tobacco industry product.
